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AERO60492 – Autonomous Mobile Robots

Lecture 3 – Revision of Linear Algebra and Probabilities

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My expectations

- You will
  - attend lectures on time
  - read instructions carefully for coursework and tutorial
  - Treat each other and myself with respect
  - Participate in various activities (such as group exercises) throughout the course
- If you need to study any async content before the lecture:
  - I will tell you explicitly over email and teams
  - The content will be available at least 5 working days in advance

Implemented changes

- Coursework timetabling
  - We have moved labs for coursework 3 one week earlier while keeping the deadline, giving you more time
  - We have moved everyone's lab to the same day
- Notations are different for the same thing across modules/fields
  - Sadly, scientific community uses different notations depending on the background, please let us know where you see this and we will try to provide clarification wherever possible

Today's lecture

- Mathematics introduction ~ 1 hour
- Exercise sheet ~ 1 hour

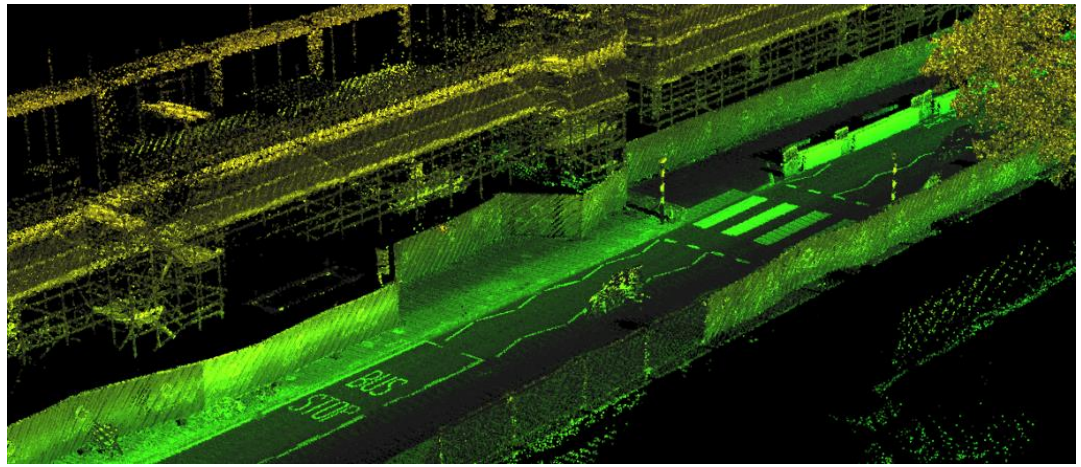
Intended learning outcomes

- To introduce basic mathematical concepts needed in this course
- To define basic operators on matrix/vectors:
  - Addition
  - Subtraction
  - dot product
  - cross product
  - elementwise multiplication
  - Transpose
  - Inverse
- To appreciate probabilities and their usage in robotics

Linear algebra relevance to robotics

- Linear algebra is present in the following aspects of robotics:
  - Robotic kinematics
  - Robot modelling
  - Sensors: vision (image is a matrix), LIDAR (for example point clouds)

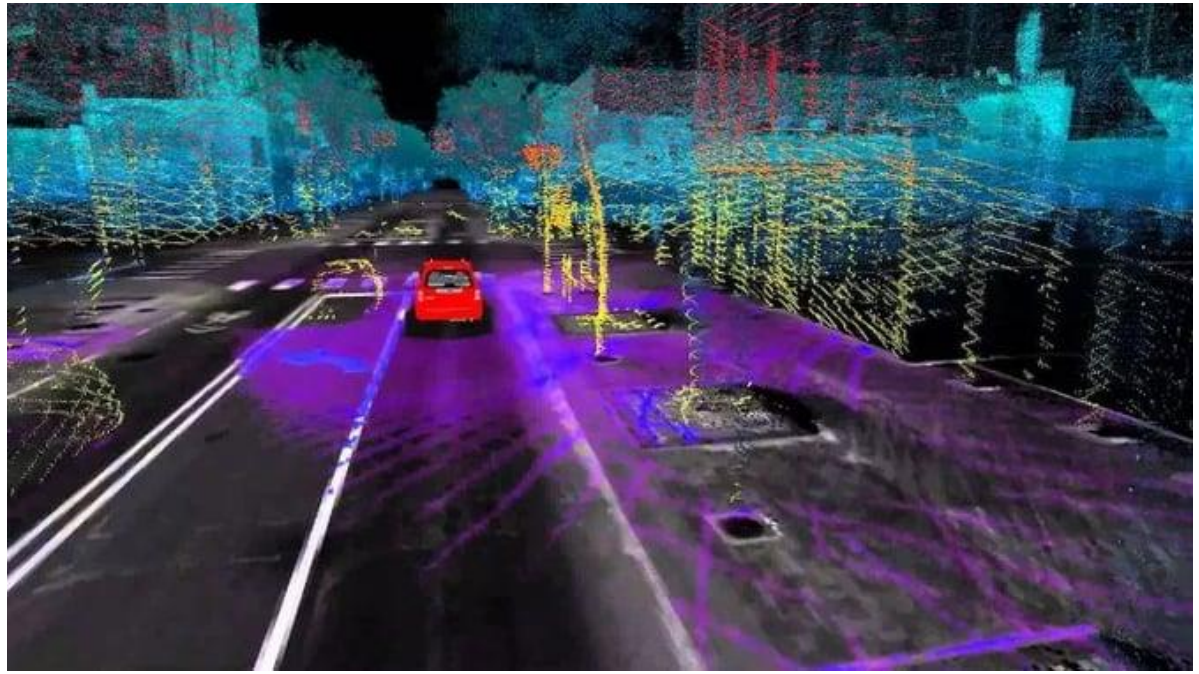
Source: https://spectrum.ieee.org/turtlebot-4



Source: https://www.oxts.com/wp-content/uploads/2020/07/pointcloud-2-banner-1024x439.png

Probability relevance to robotics

- Probabilities are used in several algorithms in robotics:
  - SLAM (including vSLAM)
  - Geospatial algorithms (i.e. how to correlate data and specific position)



Source: https://www.flyability.com/simultaneous-localization-and-mapping

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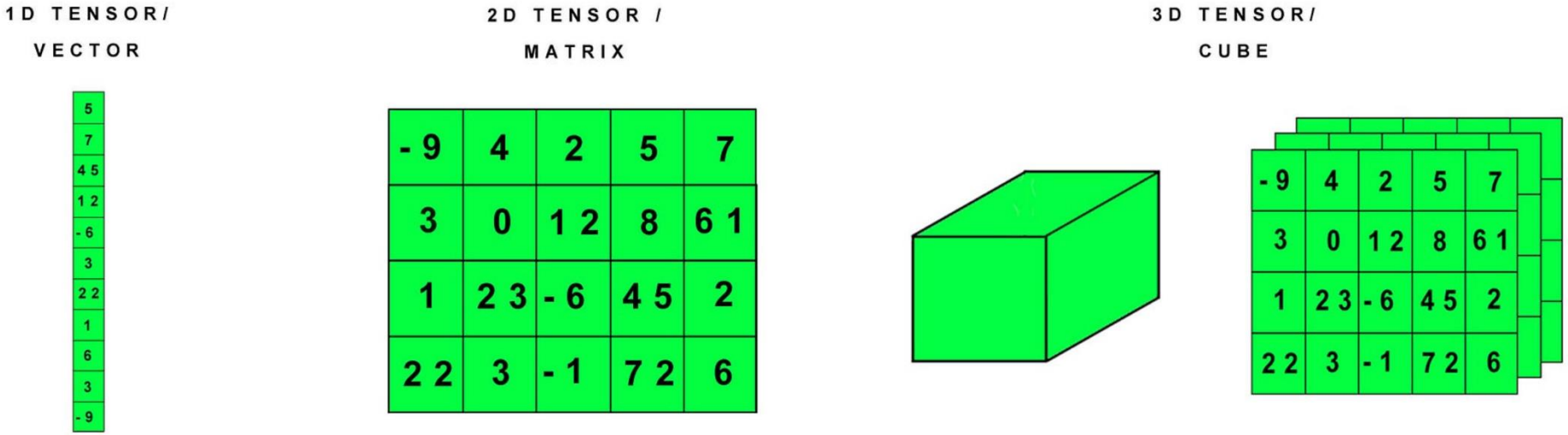
Linear Algebra

Linear algebra introduction

- Linear Algebra is a branch of mathematics which deals with any operation on vectors, matrices and tensors
- Here we only consider vectors and matrices
- vectors, matrix and tensors can be visualised as follows:



Source: https://resources.experfy.com/ai-ml/learning-ai-if-you-suck-at-math-part4-tensors-illustrated-with-cats/

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Why is it important?

- Compact representation of complex problems
- As it happens the hardware we have access to (i.e. GPUs and CPUs) can compute linear algebra equations efficiently
- You can test it yourself by doing computation in the loop vs as a vector, much faster

Notation

- There is a standard way of indicating if a given variable is a matrix, vector or a scalar
  - Scalar: denoted by small letter: v
  - Vector: denoted by boldface lowercase letter: $\boldsymbol{v}$
  - Matrix: denoted by boldface uppercase letter: $\mathbf{V}$
- There are alternative notations so be careful!

Addition/subtraction

- What is legal:
  - Adding/subtracting two tensors/vectors/matrices of same size – just add values of corresponding elements
- Adding/subtracting a scalar value:
  - Possible, just add the value to all the elements
- What is impossible:
  - Adding two matrices of different size
  - Adding vector to a matrix. Note some software package allows it by automatically expanding vector. You should **never** do this or at the very least check the result carefully.

Dot Product

- Suppose we have two matrices:

$$\mathbf{A} = \underbrace{\begin{matrix} 7 & 8 \\ 9 & 10, \\ 11 & 12 \end{matrix}}_{m} \Big\}\, n \qquad \boldsymbol{B} = \underbrace{\begin{matrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{matrix}}_{l} \Big\}\, k$$

The dot product $\boldsymbol{AB}$:

$$\boldsymbol{A} \cdot \boldsymbol{B} = \underbrace{\begin{matrix} \mathbf{7 \times 1 + 8 \times 4} & \mathbf{7 \times 2 + 8 \times 5} & \mathbf{7 \times 3 + 8 \times 6} \\ \mathbf{9 \times 1 + 10 \times 4} & \mathbf{9 \times 2 + 10 \times 5} & \mathbf{9 \times 3 + 10 \times 6} \\ \mathbf{11 \times 1 + 12 \times 4} & \mathbf{11 \times 2 + 12 \times 5} & \mathbf{11 \times 3 + 12 \times 6} \end{matrix}}_{l} \Big\}\, n$$

Easy way to compute dot product:

$$\boldsymbol{AB} = \begin{matrix} 7 & 8 \\ 9 & 10 \\ 11 & 12 \end{matrix} \cdot \begin{matrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{matrix} = \begin{matrix} & & 1 & 2 & 3 \\ & & 4 & 5 & 6 \\ \cancel{7} & \cancel{8} & 39 & 54 & 69 \\ \cancel{9} & \cancel{10} & 49 & 68 & 89 \\ \cancel{11} & \cancel{12} & 59 & 82 & 105 \end{matrix}$$

Dot product

- Properties of the dot product:
  - Only works if number of columns in $\mathbf{A}$ (m) is equal to number of rows in $\boldsymbol{B}$ (k)
  - $\boldsymbol{AB} \neq \boldsymbol{BA}$ Note: often only one of them is possible
  - Dot product is distributive: $\boldsymbol{A}(\boldsymbol{B}+\boldsymbol{C}) = \boldsymbol{AB}+\boldsymbol{AC}$
  - Associative: $\boldsymbol{A}(\boldsymbol{BC}) = (\boldsymbol{AB})\boldsymbol{C} = \boldsymbol{ABC}$
  - Note: While order does not matter using associative property for final result it matters for computational complexity. Depending on the sizes some orders are better than others!

Transpose

- Transpose flips dimensions of the vector/matrix alongside its diagonal (i.e. inverts rows and columns)
- Let's reuse matrix from previous example:

$$\mathbf{A} = \begin{matrix} 7 & 8 \\ 9 & 10 \\ 11 & 12 \end{matrix}$$

- Then $\mathbf{A}^t$:

$$\mathbf{A}^t = \begin{matrix} 7 & 9 & 11 \\ 8 & 10 & 12 \end{matrix}$$

- Note: diagonal is defined alongside the elements $a_{11}, a_{22} \dots a_{ii}$

Inverse

- The matrices cannot be divided!
- But we need a way of moving matrix from one side of equation to other, thus inverse was devised.
- Only works for some square matrices $(n \times n)$ shape
- It is formally defined as:

$$\mathbf{A}\mathbf{A}^{-1} = \boldsymbol{I}$$

Inverse

Identity matrix:

$$\begin{bmatrix} 1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 1 \end{bmatrix}$$

- Finding inverse of a matrix is non-trivial!

How do we know if matrix has an inverse?

- We can find if matrix have an inverse by computing determinant of the matrix
- Only determinant for 2×2 matrix and 3×3 matrix can be found easily.
- For higher dimensions there exist several algorithms Leibniz formula or Gaussian elimination
- Then if determinant is nonzero the matrix is invertible
- If it is non-invertible we can do some tricks to try and invert it, but this is generally beyond scope of this course

How do we find an inverse?

- Generally, we can only find inverses of small matrix by hand (such as 2×2 and 3×3)
- Regardless of the size we need the determinant – luckily, we had to find it already.
- For larger matricies algorithms is used such as Gaussian elimination, Gauss-Jordan elimination, Cholesky decomposition, QR decomposition, LU decomposition

How do we find an inverse of 2x2 matrix?

1. Find the determinant:

$$\det\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = ad - bc$$

2. Find adjugate

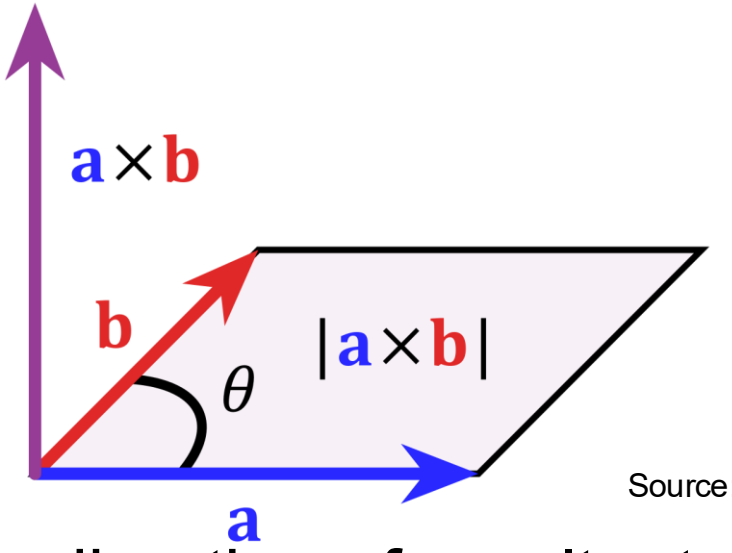
$$\operatorname{adj}\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

3. Find inverse:

$$\mathrm{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \qquad A^{-1} = \frac{1}{\det(A)}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Cross product

- Only valid for vectors
- Suppose we have two vectors: **a** = $\begin{matrix} i \\ j \\ k \end{matrix}$, **b** = $\begin{matrix} l \\ m \\ n \end{matrix}$
- Cross product results in a vector perpendicular to two vectors:



Source:https://commons.wikimedia.org/wiki/File:Cross_product_parallelogram.svg

- Note: The direction of resultant vector could be either up or down. By convention we use right-hand rule

Cross product

- Computing cross product is equivalent to computing the determinant of two vectors. In our example this is defined as follows:

$$\boldsymbol{a} \times \boldsymbol{b} = \begin{matrix} jn - km \\ kl - in \\ im - lj \end{matrix}$$

- Note: it is possible to compute cross product of arbitrary dimension vector, but it is difficult/impossible to visualise.

Element wise multiplication

- In element wise multiplication both matrices/vectors must be the same size!
- It is formally defined as Hadamard product
- It is usually denoted with $\odot$ symbol,
- **Example:**
- $\mathbf{A} = \begin{matrix} 7 & 8 \\ 9 & 10 \\ 11 & 12 \end{matrix}, \mathbf{C} = \begin{matrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{matrix}, \mathbf{A} \odot \mathbf{C} = \begin{matrix} 7 \times 1 & 8 \times 2 \\ 9 \times 3 & 10 \times 4 \\ 11 \times 5 & 12 \times 6 \end{matrix}$

Matrix operators covered:

- Addition
- Subtraction
- dot product
- cross product
- elementwise multiplication
- Transpose
- Inverse

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Probability

Introduction to probabilities

- Probability defines how likely given event is going to happen.
- Denoted by large letter P for example probability of event A is denoted as $P(A)$
- Probability can be a number, but in most robotic application it is a function
- Real world is uncertain (such as sensor errors) – probability is used to account for those
- $P(A|B)$ is called conditional probability and is defined as probability of event A happening given that B has happened

Bayes theorem general definition

- Conditional probability allows update our belief about something based on the evidence we have just seen.
- Bayes Theorem allows us to compute conditional probability.
- It is formally defined as:

Prior

Likelihood

$$P(A|B) = \frac{P(B|A) \times P(A)}{P(B)}$$

Belief (also called posterior)

Evidence

Bayes theorem in robotics

- Note: P(A) and P(B) can itself be conditional probability
- In robotics more intuitive and general definition is:

$$Posterior \propto \frac{Likelihood \times Prior}{Evidence}$$

Proportionality is used as other terms might be added depending on application

- Bayes theorem allows us to automatically account for inaccuracies!

Some key terms

- θ – hypothesis – what we are trying to determine (for example robot position). Hypothesis is often just model parameters which gets updated
- D – Data – it is either current measurement or history of measurements including current measurement
- M - Model – something (normally a function, can be several models) which models what we expect our measurements to be
- I – Prior information – information we already know and is useful for solving problem

Bayes theorem explained

- Belief/posterior $P(\theta|D,M,I)$: this is normally what we want to maximise during our mission. The higher this value the more sure we are the hypothesis is correct.
- Evidence $P(D)$: overall probability of observing the data. It is often ignored as it acts as normalizing factor
- Likelihood $P(D|\theta,M,I)$: Defines a probability of seeing a sensor reading given what we knew already, and assuming that the model and that hypothesis so far is true
- Prior $P(\theta|I)$: Express our certainty in the state of knowledge before new data point arrived

Bayes theorem high level example 1 SLAM

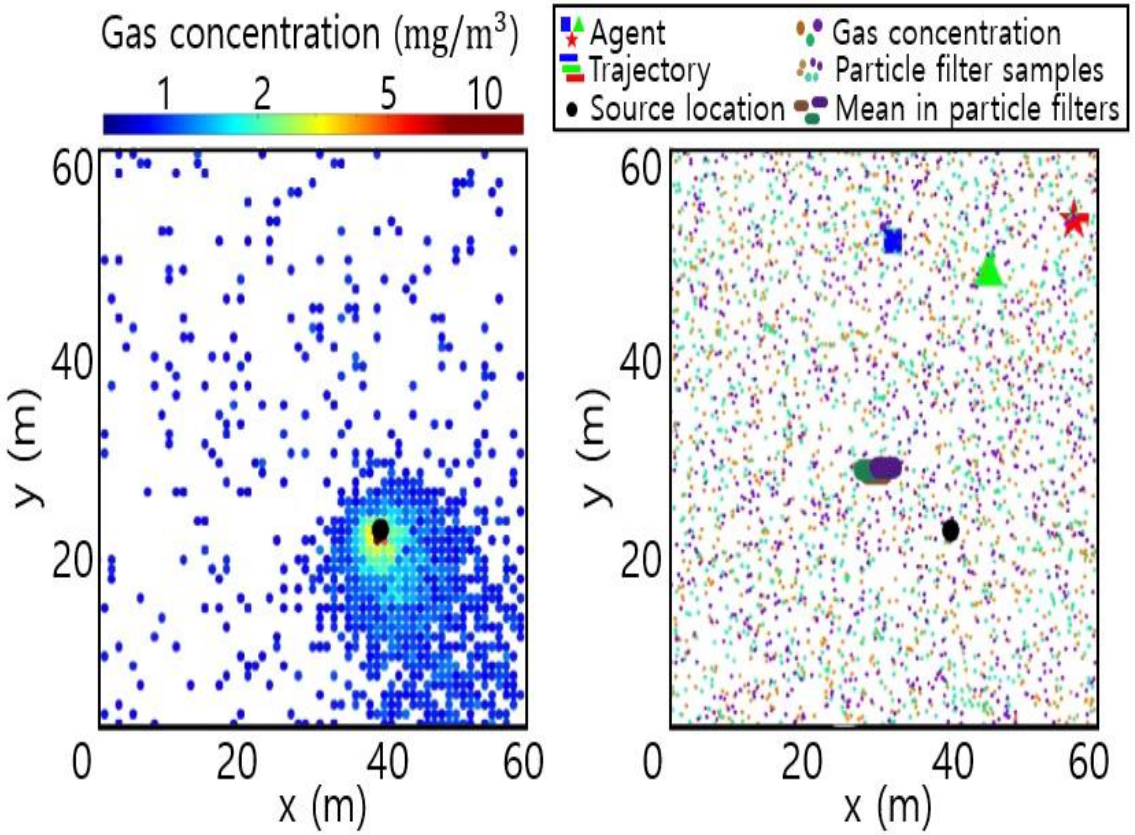
- This is a General example! It is not correct for all algorithms
  - Belief/posterior $P(\theta|D,M,I)$: How sure are we of current position given previous data, robot dynamics and seen landmarks
  - Likelihood $P(D|\theta,M,I)$: Probability of data being reasonable given where we believed robot was and the model we have
  - Prior $P(\theta|I)$: Previous step posterior

Bayes theorem high level example 2 source term estimation

- This is a General example! It is not correct for all algorithms
  - Belief/posterior $P(\theta|D,M,I)$: How sure are we of current source properties given sensors reading at specific positions and model
  - Likelihood $P(D|\theta,M,I)$: Probability of sensor data being right at this position given what we have seen already, the model and the hypothesis
  - Prior $P(\theta|I)$: Previous step posterior

Bayes theorem high level example 2 source term estimation

- Estimate properties of release of hazardous gas using a robot



Bayes theorem final thoughts

- It is very powerful tool and many problems in robotics can be fashioned after it, for example SLAM
- Normally it is solved in recursive fashion (relationship between prior and posterior from previous slides!)
- Generally, even if mathematically defined some/all needs to be approximated using other techniques such as particle filters
- Evidence is often ignored in robotics as it acts as a scaling factor only.
- Choice of likelihood, evidence and prior is arbitrary and requires a lot of trial and error and experimental backing

Summary

- In this lecture we discussed:
  - Basics of linear algebra
  - Basics of probabilities
  - Introduction to bayes theorem
  - Relevance of both to robotics

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Thank you for listening